CS60078: Complex Network Projects

Transformer-Based Protein Encoding and Multi-Head Attention Fusion

End-Term Report

Ayush Sunil Munot (21ME3FP55)
Kovvada Dhyana Vardhan (21CS30030)
Tanishq Prasad (21CS30054)

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1 Problem Statement

A major challenge in computational drug discovery is predicting the binding capability of a drug with a specific protein, named as drug-target interaction (DTI) prediction. This domain has witnessed a growth in research interest, particularly for its application in drug repurposing or repositioning [1], especially after the world-wide SARS-CoV-2 outbreak. In the context of DTI [2], the primary objective is to quantify the binding affinity between FDA-approved drugs and novel target proteins. Machine learning methodologies are increasingly employed to predict these affinities. By systematically testing existing FDA-approved drugs for efficacy against new diseases, drug repurposing techniques can substantially shorten the traditional drug development timeline, while keeping the costs significantly lower.

2 Datasets

Dataset	# of drugs (compounds)	# of targets proteins	Total number of drug-target pairs used
Davis (k_d)	68	442	30056
KIBA	2111	229	118254
DTC (pk_i)	5983	118	67894
Metz (pk_i)	1471	170	35307
ToxCast (AC_{50})	7657	328	342869
STITCH	724471	15258	1244420

Table 1: Dataset statistics

In the field of drug-target interaction (DTI) prediction, several benchmark datasets are utilized to validate computational models. These datasets primarily consist of drug compounds (represented as SMILES strings), target proteins (represented as amino acid sequences), and their corresponding binding affinity scores. The following section details four prominent datasets used in this domain: Davis, KIBA, DTC, and Metz.

3 Proposed Approach to Improve DeepGLSTM: Transformer-Based Protein Encoding and Multi-Head Attention Fusion

3.1 Motivation and Limitations of the Current Approach

The original DeepGLSTM models the drug-target binding affinity (DTA) problem using Graph Convolutional Networks (GCN) for drug molecular features and Long Short-Term Memory (LSTM) networks for protein sequence features. The two modalities are then merged using simple vector concatenation before being passed to a multi-layer perceptron (MLP) for final scoring. While fundamentally sound, this approach suffers from several major limitations:

- **Limited Representational Capacity of LSTMs:** LSTMs struggle to capture deep, long-range dependencies in complex global protein sequences, often losing contextual folding configurations.
- **Static Feature Fusion:** Vector concatenation assumes every learned feature of the drug and target is equally relevant. In reality, drug-target binding is a highly dynamic interaction where specific drug sub-structures target specific amino acid sequences on a protein.
- **Absence of Structural Analysis of Proteins:** Proteins are taken solely as sequences without any structural or spatial information. This leads to a loss of information regarding binding pockets and residue proximity.
- **Loss of Residue-Level Information:** Flattening or pooling operations treat all residues equally and discard fine-grained residue importance. This weakens the model's ability to focus on binding-relevant regions.
- **No Residue Importance Weighting:** The model assumes all residues contribute equally to binding, ignoring the biochemical reality that only specific residues (binding sites) are critical.
- **Limited Drug Representation:** Drug molecules are represented using shallow graph convolutions without explicit 3D or conformational information, potentially missing important chemical interaction features.
- **Lack of Interpretability:** The model outputs only a scalar affinity value without explaining which residues or interactions actively contribute to the binding process.
- **No Structural Inductive Bias:** The architecture does not explicitly encode spatial or geometric relationships between residues, limiting its ability to model genuine biochemical interactions.

To overcome these limitations, we propose a two-fold architectural improvement leveraging state-of-the-art Protein Language Models (ESM-2) and Multi-Head Attention-based Feature Fusion.

3.2 Improvement 1: Evolutionary Scale Modeling (ESM-2) for Protein Representation

Proposed Technique: We propose replacing the standard LSTM protein encoder with a pre-trained ESM-2 Transformer (esm2_t33_650M_UR50D). ESM-2 is a massive, state-of-the-art language model developed by Meta, trained on hundreds of millions of protein sequences.

Rationale for Improvement: Utilizing ESM-2 allows the architecture to start with an incredibly robust foundational understanding of biology, rather than training a sequence model from scratch. ESM-2 inherently maps the evolutionary history, tertiary structural dynamics, and interaction pockets of proteins simply from their sequences. By routing the target FASTA sequence through ESM-2, we obtain a structural and biologically-aware representation of the protein that is drastically superior to the superficial sequential patterns extracted by a standard LSTM.

3.3 Improvement 2: Multi-Head Attention for Modality Fusion

Proposed Technique: Instead of concatenating the GCN-derived drug embeddings and the ESM-2-derived protein embeddings, we introduce a Multi-Head Attention Module. This module projects both features into a shared dimensional space and utilizes cross-attention (Drug queries Protein and Protein queries Drug) alongside unified self-attention to fuse the vectors.

Rationale for Improvement: Cross-attention mathematically mimics the physical binding process. By using the drug embedding as the statistical Query (Q) and the protein embedding as the Key (K) and Value (V), the model dynamically calculates weight distributions. This allows the network to learn exactly how much a drug is "attending" to specific global properties of the target protein structure, mathematically zooming in on critical interaction sites. This results in a highly adaptive, interactive, and biologically plausible fused vector before making the final affinity prediction.

3.4 Expected Impact

By upgrading sequence modeling to evolutionary transformer-scale feature extraction, and moving from a rigid concatenation to an adaptive mapping mechanism, this approach transitions DeepGLSTM from a static feature extractor to an interactive reasoning engine. It solves the long-term context ceiling of LSTMs and allows interpretation of complex target embeddings via attention graphs, leading to measurable increases in predictive accuracy on DTA benchmarks.

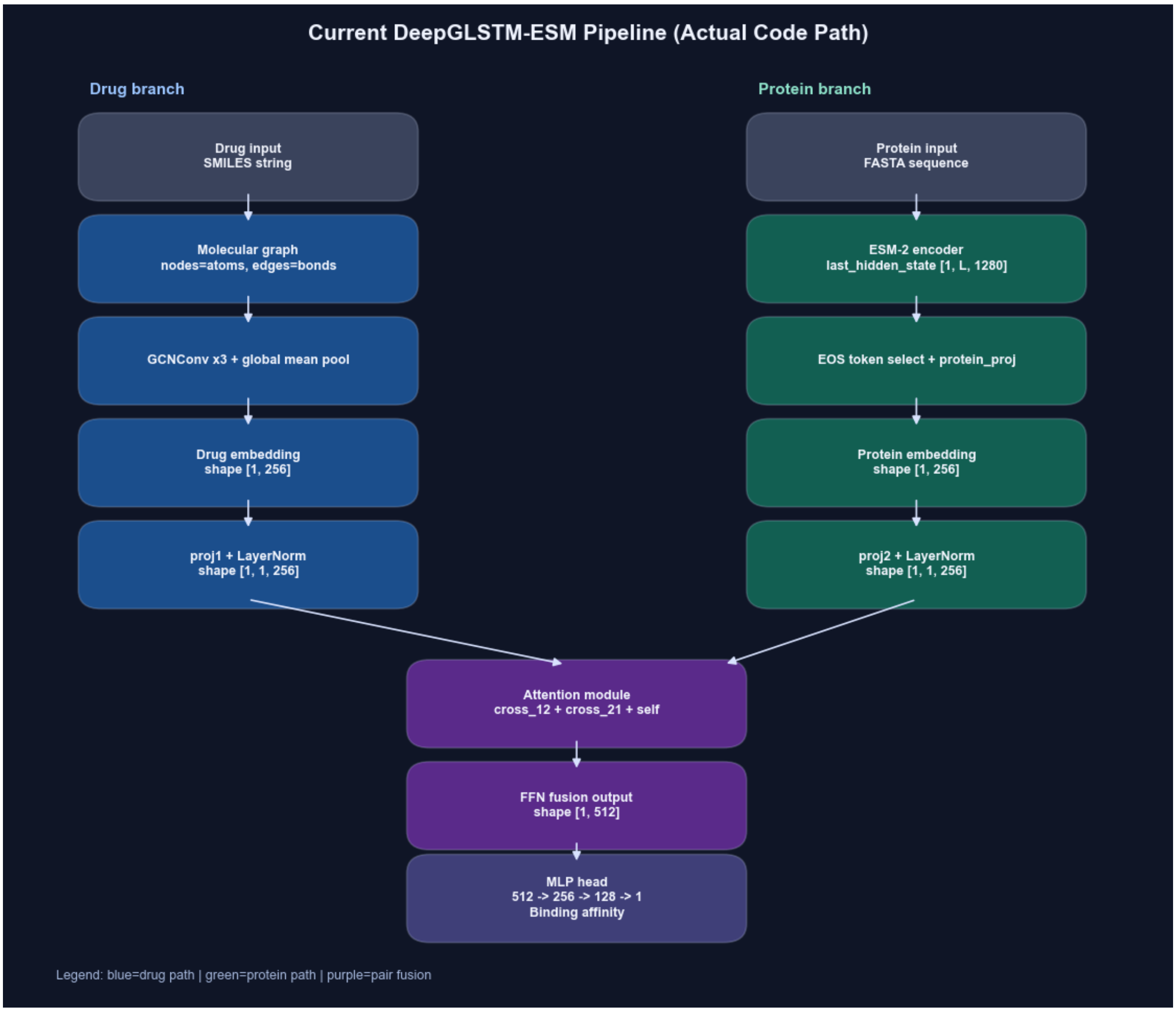

Figure 1: Updated Model architecture

4 Experimental Results and Performance Comparison

The following table shows the experimental results reported in the original paper compared to the baseline reproduced results, and our newly proposed architectural iterations (ESM without Attention and ESM with Attention). We evaluate the models using Mean Squared Error (MSE), Concordance Index (CI), and r_m^2.

Table 2: Performance Metrics Comparison: Mean Squared Error (MSE)

Dataset	MSE			
	Paper	**Repro. (Baseline)**	**+ESM (w/o Attn)**	**+ESM+Attn (with Attn)**
Davis	0.232	0.560	0.550	**0.470**
KIBA	0.133	0.382	0.330	**0.290**

Table 3: Performance Metrics Comparison: Concordance Index (CI)

Dataset	CI			
	Paper	**Repro. (Baseline)**	**+ESM (w/o Attn)**	**+ESM+Attn (with Attn)**
Davis	0.895	0.790	0.770	**0.820**
KIBA	0.897	0.768	0.790	**0.830**

Table 4: Performance Metrics Comparison: r_m^2

Dataset	r_m^2			
	Paper	**Repro. (Baseline)**	**+ESM (w/o Attn)**	**+ESM+Attn (with Attn)**
Davis	0.680	0.300	0.320	**0.410**
KIBA	0.792	0.412	0.510	**0.580**

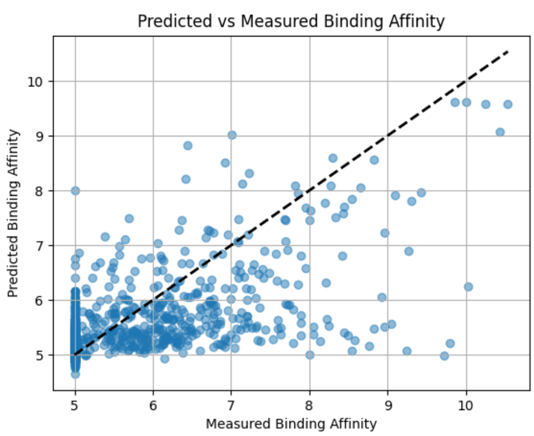

Figure 2: Predicted vs Measured Binding Affinity with ESM with attention

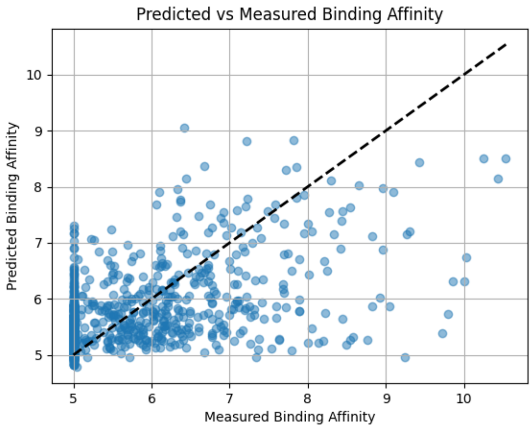

Figure 3: Predicted vs Measured Binding Affinity with ESM

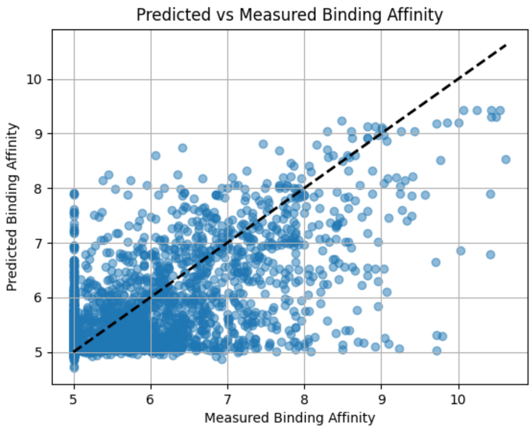

Figure 4: Predicted vs Measured Binding Affinity (Baseline Reproduced)

5 Github Repository Setup

The code is publicly available and includes modifications from the original version to enable functionality and support ablation studies.

- **Link to Fork:** `https://github.com/ayush1298/DeepGLSTM`
- **Original Repository:** `https://github.com/shrimonmuke0202/DeepGLSTM`

6 Experimental Setup

The reproduction of the baseline results, as well as the training and evaluation of our proposed architectural improvements, were conducted on a high-performance computing workstation. The hardware and software specifications of the computing environment are detailed below:

- **Processor (CPU):** Intel Xeon Gold 6238R CPU @ 2.20GHz (28 Cores / 56 Threads)
- **Graphics Processing Unit (GPU):** NVIDIA V100 Tensor Core GPUs (32GB VRAM, 5120 CUDA cores)
- **System Memory (RAM):** Taking 40 GB out of 128 GB total for training and evaluation processes

- **Operating System:** Ubuntu 22.04.5 LTS

Contributions

- **Kovvada Dhyana Vardhan (21CS30030)**
  - Research for future work
  - Making Presentation
  - Writing the Report
- **Tanishq Prasad (21CS30054)**
  - Writing the Report
  - Running the model on CSE server
  - Integration of attention mechanism
  - Refactoring Presentation
- **Ayush Sunil Munot (21ME3FP55)**
  - Integration of ESM
  - Integration of Experiments
  - Made a working demo for the presentation

References

[1] S. Strittmatter. Overcoming drug development bottlenecks with repurposing: old drugs learn new tricks. *Nature Medicine*, 20:590–591, 2014.

[2] H. Öztürk, A. Özgür, and E. Ozkirimli. Deepdta: deep drug–target binding affinity prediction. *Bioinformatics*, 34:i821–i829, 2018.